THE SOUTH AFRICAN MATHEMATICS OLYMPIAD

Senior Second Round 2024

Solutions

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1. **Answer 025**

   Adding the first equation to twice the second equation gives $3a + 20 + 4a = 55$, so $7a = 35$ and $a = 5$. Then $b = 10 + 2 \times 5 = 20$, so $a + b = 25$.

2. **Answer 504**

   If $0 < a < b$, then $\frac{a+1}{b+1} - \frac{a}{b} = \frac{b-a}{b(b+1)} > 0$, so $\frac{a+1}{b+1} > \frac{a}{b}$. Thus the largest fraction is the last one, with its numerator equal to 504.

3. **Answer 054**

   If the area of the sector is 15% of the area of the whole circle, then the angle at the centre is 15% of a full revolution, 360°. Thus $x^\circ = 15\% \times 360^\circ = 54^\circ$.

4. **Answer 002**

   Since a and b are both positive, we can write $\frac{a}{b} = x^2$. Then $\frac{a}{b} + \frac{b}{a} = x^2 + x^{-2} = (x - x^{-1})^2 + 2 \geq 2$, since a perfect square cannot be negative. The value 2 is attained when $a = b$.

5. **Answer 080**

   If the bottom left angle, in degrees, is equal to y, then the other two unmarked angles are also equal to y, because of the two isosceles triangles. Thus $180 = 3y + 30$, using the angle-sum of the whole triangle, so $y = 50$. Then $x = 180 - 2y = 180 - 100 = 80$ (angle-sum) or $x = y + 30 = 80$ (exterior angle).

6. **Answer 007**

   $2^{2024} \times 5^{2026} = 10^{2024} \times 5^2$, which is written out as 25 followed by 2024 zeros. The sum of the digits is therefore $2 + 5 = 7$.

7. **Answer 081**

$$\begin{aligned}\sqrt[3]{\left(25+\sqrt[4]{14+\sqrt{2+\sqrt[3]{8}}}\right)^4} &= \sqrt[3]{\left(25+\sqrt[4]{14+\sqrt{2+2}}\right)^4}\\ &= \sqrt[3]{\left(25+\sqrt[4]{14+2}\right)^4}\\ &= \sqrt[3]{(25+2)^4}\\ &= \sqrt[3]{27^4}\\ &= \left(\sqrt[3]{27}\right)^4\\ &= 81\end{aligned}$$

8. **Answer 008**

For $r = 1, \ldots, 23$, Martian r will call out all numbers with remainder r after division by 24, and Martian 24 will call out the numbers exactly divisible by 24. Since $2024 = 84 \times 24 + 8$, it follows that 2024 will be called out by Martian 8.

9. **Answer 016**

We need to find the highest power of 2 dividing each of the even numbers from 2 to 18 and add them up. The answer is $1 + 2 + 1 + 3 + 1 + 2 + 1 + 4 + 1 = 16$.

10. **Answer 005**

If there are $2k$ people in the row, then after k moves the even numbers have dropped out, and the row is $1, 3, \ldots, (2k-1)$. If there are $2k + 1$ people to start with, then again after k moves we are left with $(2k+1), 1, 3, \ldots, (2k-1)$. Starting with 10 people, after five moves we have $1, 3, 5, 7, 9$, after two more we have $9, 1, 5$, then $5, 9$, then 5.

11. **Answer 061**

If A is at the origin $(0, 0)$, then C is at $(9, 10)$, and coming back diagonally, G is at $(7, 8)$ and K is at $(5, 6)$. Thus $AK^2 = 5^2 + 6^2 = 61$.

12. **Answer 144**

Since y is a factor of both 9 and 16, which have highest common factor 1, it follows that $y = 1$, so $xw = 9 \times 16 = 144$.

13. **Answer 872**

The number of ways of arranging n objects in a line is $n\times(n–1)\times\cdots\times3\times2\times1$, which is denoted $n!$ (or in words n factorial). From 08:00 till 13:00 means the shopkeeper works for 5 hours. Thus the total number of arrangements made by the shopkeeper is $6! + 5! + 4! + 3! + 2! = 720 + 120 + 24 + 6 + 2 = 872$.

14. **Answer 004**

$99\,999^2 = (10^5 - 1)^2 = 10^{10} - 2.10^5 + 1 = 10\,000\,000\,001 - 200\,000 = 9\,999\,800\,001$.

15. **Answer 060**

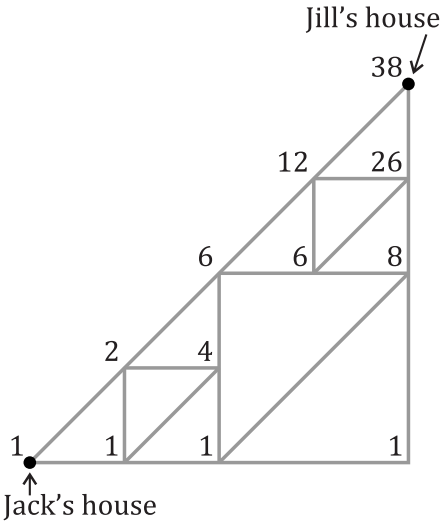
It takes 4 seconds for the whole train to pass the end of the platform, leaving $9 - 4 = 5$ seconds for the front of the train to travel from the beginning to the end of the platform. The train's speed is therefore $75/5 = 15$ m/s, and its length is $4 \times 15 = 60$ m.

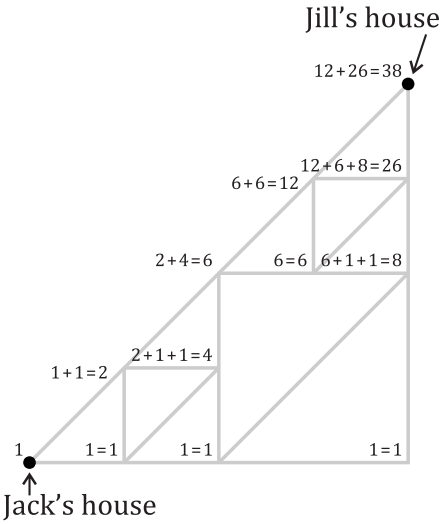
16. **Answer 260**

Since $(x \pm y)^3 = x^3 \pm 3x^2y + 3xy^2 \pm y^3$, it follows that $(x+y)^3 = 28+26 = 54 = 3^3.2$ and $(x - y)^3 = 28 - 26 = 2$. Thus $x + y = 3.2^{1/3}$ and $x - y = 2^{1/3}$, giving $x = 2.2^{1/3} = 2^{4/3}$ and $y = 2^{1/3}$. Finally, $x^6 + y^6 = 2^8 + 2^2 = 256 + 4 = 260$.

17. **Answer 038**

The number of ways to each node (where two or more paths join) is equal to the sum of the number of ways to the nodes immediately to the West, South-West or South of it. Thus we can draw up the following diagrams, from bottom left to top right.





18. **Answer 465**

Since $f(2) = 2+f(1) = 2+1$ and $f(3) = 3+f(2) = 3+2+1$, and so on, we see that $f(30) = 30+29+28+\cdots+3+2+1 = 1+2+3+\cdots+28+29+30$. By adding the two sums together term by term, we see that $2f(30) = 31+31+\cdots+31+31 = 30\times 31$, so $f(30) = 15 \times 31 = 465$.

(Alternatively, use the formula for the sum of an arithmetic series.)

19. **Answer 050**

If O is the centre of the circle, then the shaded region is obtained by taking the semicircle with diameter AB and the triangle OAB, and then removing the quarter-circle OAB. The quarter-circle has radius 10, and by Pythagoras' theorem the semicircle has radius $5\sqrt{2}$. Thus the shaded area is $\frac{1}{2}\pi(5\sqrt{2})^2 + (5\sqrt{2})^2 - \frac{1}{4}\pi(10)^2 = 25\pi + 50 - 25\pi = 50$.

20. **Answer 064**

If Burgerbraai has n toppings available, then 2^n selections can be made (from no toppings to all toppings), since each topping is either on or off. Rocoburger has $(n+2)$ toppings and therefore 2^{n+2} selections. Thus $2^{n+2} - 2^n = 192$, giving $2^n(2^2-1) = 192$, so $2^n = 192 \div 3 = 64$ and $n = 6$.

21. **Answer 024**

If $FB = x$, then $FC = x+6$. Triangles EBF and EAG are congruent (AAS) so $EG = FE = \sqrt{x^2+9}$, giving $FG = 2\sqrt{x^2+9}$. Since triangle FC is isosceles, it follows that $2\sqrt{x^2+9} = x+6$, and squaring gives $4(x^2+9) = (x+6)^2$. This simplifies to $3x^2 = 12x$ so $x = 4$, since $x > 0$. The shaded region is obtained by removing triangles AEG and CDG from the square $ABCD$, so its area is equal to $6^2 - \frac{1}{2}(3)(4) - \frac{1}{2}(2)(6) = 24$.

22. **Answer 840**

We begin with the identity $\frac{1}{k^2+k} = \frac{1}{k} - \frac{1}{k+1}$. Thus, we have

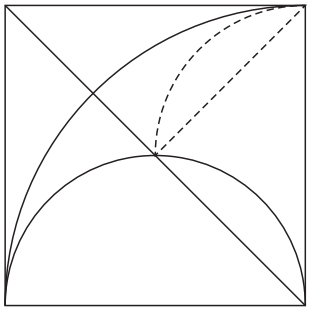
$$\begin{aligned} a_m + a_{m+1} + \cdots + a_{n-1} &= \left(\tfrac{1}{m} - \tfrac{1}{m+1}\right) + \left(\tfrac{1}{m+1} - \tfrac{1}{m+2}\right) + \cdots + \left(\tfrac{1}{n-1} - \tfrac{1}{n}\right) \\ &= \tfrac{1}{m} - \tfrac{1}{n} \\ &= \tfrac{1}{29} \\ \Rightarrow mn + 29m - 29n &= 0 \\ \Rightarrow (m-29)(n+29) &= -29^2 \end{aligned}$$

We deduce that $m - 29 = -1$ and $n + 29 = 29^2$. Thus, $m + n = 840$.

23. **Answer 242**

Triangles ABC and ABD have the same area, since they have the same base and the same height, from which it follows that triangles APD and BPC have the same area, say x. If h and k denote the perpendicular distances from P to AB and CD respectively, then $\triangle ABC = \frac{1}{2}(AB)(h+k)$ and $\triangle ABP = \frac{1}{2}(AB)h$. Thus $\frac{h+k}{h} = \frac{72+x}{72}$, giving $\frac{k}{h} = \frac{x}{72}$. Similarly, $\frac{h}{k} = \frac{x}{50}$, and multiplying the two equations gives $1 = \frac{x}{72}\frac{x}{50}$. Thus $x^2 = 72 \times 50 = 3600$, so $x = 60$. The total area of the trapezium is $72 + 50 + 2x = 122 + 2 \times 60 = 242$.

24. **Answer 006**



From the sketch of half of the rectangle, it is clear that the three originally white regions can be moved to cover exactly one-quarter of the square without overlapping. The same applies to the other half of the rectangle, so the shaded regions in both halves occupy three-quarters of the rectangle, and their total area is $\frac{3}{4} \times 4 \times 2 = 6$.

25. **Answer 006**

By adding $2a + 1$ to both sides of the equation, we obtain $(a + 1)^2 - b^2 = 2025$, so $(a + 1 + b)(a + 1 - b) = 2025$. Since $2025 = 3^4 \cdot 5^2$, it has $(4 + 1)(2 + 1) = 15$ factors, which give seven pairs (d, q) with $q = \frac{2025}{d} > d > 0$ and one factorization with $q = d = 45$. Each of the seven factorizations $2025 = q \times d$ with $q > d > 0$ gives $a + 1 + b = q$ and $a + 1 - b = d$, so $a = \frac{1}{2}(q + d) - 1$ and $b = \frac{1}{2}(q - d)$, which are positive integers since q and d are both odd and $q > d$. However, the question requires two different positive integers, which is not the case for $d = 1$ and $q = 2025$, that yields $a = b = 1012$. Therefore, 6 number pairs remain.

d	1	3	5	9	15	25	27	45
q	2025	675	405	225	135	81	75	45